

Firebase Realtime Database

1. AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.PML03RealtimeDB"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="
                    android.intent.action.MAIN" />
                <category android:name="
                    android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

2. build.gradle (Project)

```
// Top-level build file where you can add configuration options common to
all sub-projects/modules.
plugins {
    alias(libs.plugins.android.application) apply false
    alias(libs.plugins.kotlin.android) apply false
    id("com.google.gms.google-services") version "4.4.3" apply false
}
```

3. build.gradle (module)

```
plugins {
    alias(libs.plugins.android.application)
    alias(libs.plugins.kotlin.android)
    id("com.google.gms.google-services")
}
```

```

}

android {
    namespace = "com.mi.pml03realtimedb"
    compileSdk = 36

    defaultConfig {
        applicationId = "com.mi.pml03realtimedb"
        minSdk = 25
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
            "androidx.test.runner.AndroidJUnitRunner"
    }

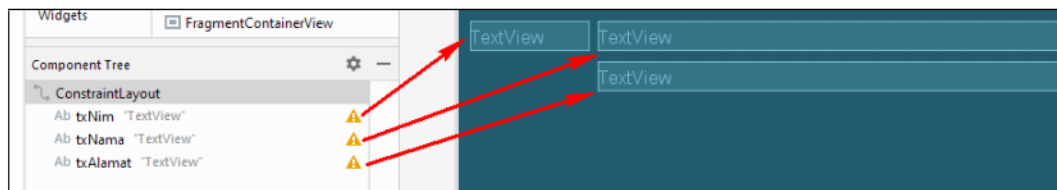
    buildFeatures {
        viewBinding = true
    }

    buildTypes {
        release {
            isMinifyEnabled = false
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
                "proguard-rules.pro"
            )
        }
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
    kotlinOptions {
        jvmTarget = "11"
    }
}

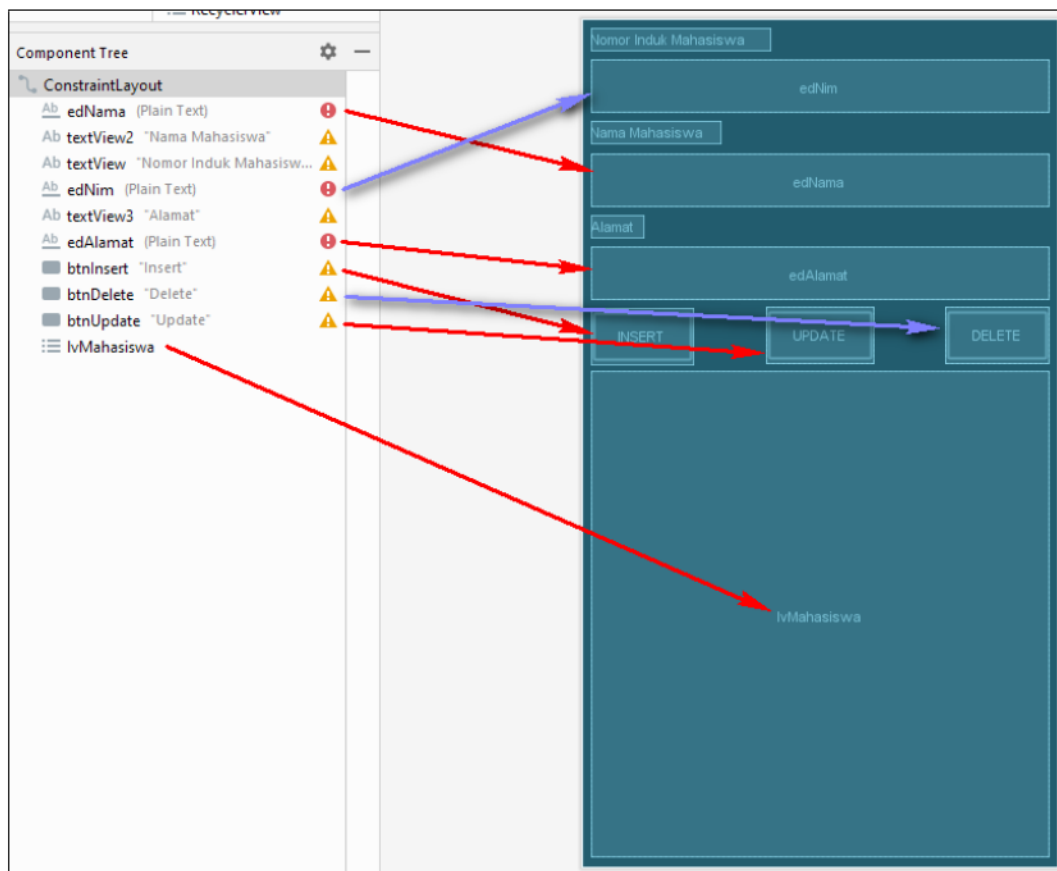
dependencies {
    implementation(platform("com.google.firebase:firebase-bom:34.2.0"))
    implementation("com.google.firebase:firebase-database")
    implementation(libs.androidx.core.ktx)
    implementation(libs.androidx.appcompat)
    implementation(libs.material)
    implementation(libs.androidx.activity)
    implementation(libs.androidx.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.androidx.junit)
    androidTestImplementation(libs.androidx.espresso.core)
}

```

4. row_mahasiswa.xml



5. activity_main.xml



6. Mahasiswa.kt (class)

```
package com.mi.pml03realtimeadb

class Mahasiswa {
    var namaMhs: String? = null
    var alamatMhs: String? = null
    var nim: String? = null
}
```

7. MainActivity.kt (class)

```
package com.mi.pml03realtimedb

import android.os.Bundle
import android.view.View
import android.widget.AdapterView
import android.widget.ListAdapter
import android.widget.SimpleAdapter
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
import com.google.firebase.database.*
import com.mi.pml03realtimedb.databinding.ActivityMainBinding

class MainActivity : AppCompatActivity(), View.OnClickListener {

    lateinit var db : DatabaseReference
    lateinit var adapter : ListAdapter
    var alMhs = ArrayList<HashMap<String,Any>>()
    var mhs = Mahasiswa()
    var hm = HashMap<String,Any>()
    lateinit var b : ActivityMainBinding

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        b = ActivityMainBinding.inflate(layoutInflater)
        enableEdgeToEdge()
        setContentView(b.root)
        ViewCompat.setOnApplyWindowInsetsListener(
            findViewById(R.id.main)) { v, insets ->
                val systemBars =
                    insets.getInsets(WindowInsetsCompat.Type.systemBars())
                v.setPadding(systemBars.left, systemBars.top,
                    systemBars.right, systemBars.bottom)
                insets
            }

        FirebaseDatabase.getInstance().setPersistenceEnabled(true)
        b.btnInsert.setOnClickListener(this)
        b.btnDelete.setOnClickListener(this)
        b.btnUpdate.setOnClickListener(this)
        b.lvMahasiswa.setOnItemClickListener(itemClickListener)
    }

    override fun onStart() {
        super.onStart()
        db =
            FirebaseDatabase.getInstance().getReference("TabelMahasiswa")
        showData()
    }
}
```

```

fun showData(){
    db.addValueEventListener(object : ValueEventListener{
        override fun onDataChange(snapshot: DataSnapshot) {

            val dataSnapshotIterable = snapshot.children
            val iterator = dataSnapshotIterable.iterator()
            alMhs.clear()
            while (iterator.hasNext()){
                mhs = iterator.next().getValue<Mahasiswa>()!!
                hm = HashMap()
                hm.put("nim", mhs.nim!!)
                hm.put("namaMhs", mhs.namaMhs!!)
                hm.put("alamatMhs", mhs.alamatMhs!!)
                alMhs.add(hm)
            }

            //binding data didalam arraylist dengan layout
            //row_mahasiswa.xml
            //untuk ditampilkan melalui listview
            adapter = SimpleAdapter(
                this@MainActivity,
                alMhs,
                R.layout.row_mahasiswa,
                arrayOf("nim", "namaMhs", "alamatMhs"),
                intArrayOf(R.id.txNim, R.id.txNama, R.id.txAlamat)
            )
            b.lvMahasiswa.setAdapter(adapter)
        }

        override fun onCancelled(error: DatabaseError) {
            Toast.makeText(this@MainActivity,
                "Connection to database error : ${error.message}",
                Toast.LENGTH_SHORT).show()
        }
    })
}

override fun onClick(p0: View?) {
    when(p0!!.id){
        R.id.btnInsert -> {
            //Toast.makeText(this, "hello world",
            //Toast.LENGTH_SHORT).show()
            mhs.nim = b.edNim.text.toString()
            mhs.namaMhs = b.edNama.text.toString()
            mhs.alamatMhs = b.edAlamat.text.toString()
            db.child(mhs.nim!!).setValue(mhs)
        }
        R.id.btnDelete -> {
            db.child(b.edNim.text.toString()).removeValue()
        }
        R.id.btnUpdate -> {
            db.child(mhs.nim!!).child("namaMhs")

```

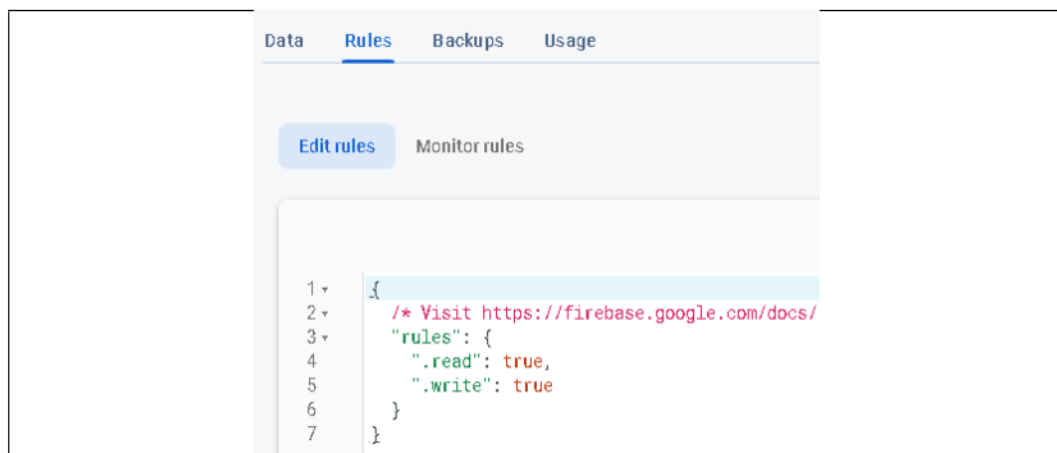
```

        .setValue(b.edNama.text.toString())
        db.child(mhs.nim!!).child("alamatMhs")
        .setValue(b.edAlamat.text.toString())
    }
}
b.edNim.setText(""); b.edNama.setText(""); b.edAlamat.setText("")
}

val itemClickListener = AdapterView.OnItemClickListener {
    adapterView, view, i, l ->
    hm = HashMap()
    hm = alMhs.get(i)
    b.edNim.setText(hm.get("nim").toString())
    b.edNama.setText(hm.get("namaMhs").toString())
    b.edAlamat.setText(hm.get("alamatMhs").toString())
}
}

```

8. Firebase Realtime Database Rules



9. Firebase Project Setting

